U.9: BisectionEnsemble: Integrating Local Feasibility Sampling into the Redistricting Bisection Tree

Gio Della-Libera

Draft — May 12, 2026

Abstract

Markov-chain redistricting ensemble methods (ReCom, GerryChain) sample the full plan space by running k-way recombination steps, but exhibit catastrophic bipartition failure at large k: Pennsylvania ($k = 17$, prime) requires a 17-way bipartition of the whole state; Texas ($k = 38 = 2 \times 19$) requires a 19-way bipartition at the first ApportionRegions split. These large-arity bipartitions have no efficient graph-theoretic decomposition and cause GerryChain cold-start failures. We introduce *BisectionEnsemble*, which resolves this failure by repositioning ensemble sampling inside the bisection tree rather than around it. At each node of the bisection tree—which decomposes a k-district state into a hierarchy of 2-way cuts—we run a local T-step ReCom chain on the $\approx n/k$-tract subregion and select the bisection cut at percentile p of the collected edge-cut distribution. Since every local ensemble is always 2-way ($k_{\text{local}} = 2$), bipartition failure at the local ReCom level is structurally impossible. The bisection tree structure (ApportionRegions, GeoSection) is fully preserved. We implement `SeedCompositor::BisectionEnsemble{p, ensemble_steps}` in `redist-cli` using `split_subgraph_bisection_ensemble()`, which seeds a METIS bisection as the Markov chain's initial state, runs `ensemble_steps` ReCom steps via `redist_ensemble::recom::RecomChain`, collects accepted plans with their edge cuts, and returns the plan at rank $\lfloor p \times \text{accepted\_count} \rfloor$. Empirical evaluation on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$) shows that BisectionEnsemble(p=0.5, T=100) improves Polsby-Popper compactness by 5–12% over standard METIS bisection, matches or improves partisan proportionality, and crucially, succeeds on TX where direct GerryChain $k = 38$ fails with bipartition errors. Runtime is $O(T \cdot n \log n)$ total and trivially parallelisable across tree nodes.

1 Introduction

Redistricting ensemble methods have emerged as a principled approach to evaluating and generating congressional maps. By sampling from the space of all legally-feasible plans, methods such as ReCom [DeFord et al., 2021] and GerryChain [Duchin and Walch, 2019] provide a reference distribution against which any proposed map can be measured for partisan or racial bias. The core primitive is a *recombination step*: two adjacent districts are merged and re-split along a random spanning tree, producing a new feasible plan in $O(n)$ time per step.

The bipartition failure problem. This approach works well when the number of districts k is small or when k decomposes into small factors. But many states have seat counts that are large primes or contain large prime factors: Texas has $k = 38 = 2 \times 19$, Pennsylvania has $k = 17$ (prime), New York has $k = 26 = 2 \times 13$, and Michigan has $k = 13$ (prime). A direct k-way ReCom ensemble on such states requires, at each step, merging two adjacent districts and re-splitting the merged subgraph into a balanced bipartition. The correct failure mode is geometric, not a balance-ratio issue: in a $k = 38$ state, merging two adjacent districts produces a region whose spanning tree must achieve a $1 : 1$ balance *within* the merged region, but the surrounding $k - 2 = 36$ districts constrain the boundary so severely that balanced spanning tree cuts become rare. GerryChain *without pair reselection* stalls in this regime; GerryChain *with pair reselection* can succeed by trying different adjacent pairs until a balanced cut is found, at the cost of lower throughput in Python.

BisectionEnsemble: local 2-way ReCom at every node. We propose a structural fix: instead of running a k-way ensemble at the top level, embed 2-way ensemble sampling *inside* the bisection tree. The bisection tree (see ApportionRegions [Della-Libera, 2026a] and GeoSection [Della-Libera, 2026d]) already decomposes the redistricting problem into a hierarchy of 2-way cuts. At each node of this tree—which manages a subregion of $\approx n/\text{level}$ tracts and must split it into 2 balanced parts—we replace the single METIS call with a T-step local ReCom chain seeded by the METIS bisection.

Because every local ensemble is always 2-way at bisection nodes, bipartition failure at the local ReCom level is structurally impossible: we never ask for a $1 : 37$ balance at the full-state level, only ever a $1 : 1$ balance within each node's subregion (or near-$1 : 1$ for prime seat counts using the floor/ceil split). The bipartition failure guarantee applies to the local ReCom steps at each bisection node; METIS calls at $p' > 2$ ApportionRegions nodes are not covered by this guarantee. The bisection tree structure is fully preserved, guaranteeing population balance and contiguity through the same mechanisms as the original bisection algorithm.

Key advantages.

1. **Always tractable at bisection nodes**: No bipartition failures at the local ReCom level regardless of k. TX ($k = 38$), PA ($k = 17$), NY ($k = 26$) all succeed at their bisection nodes.

2. **Local feasibility sampling**: Each node samples the local feasible space of bisections of its subregion. At depth ℓ, each node manages $\approx n/2^{\ell}$ tracts; the locality advantage grows with depth. The root node ($\ell = 0$) manages half the state, while leaf nodes manage $\approx n/k$ tracts.

3. **Parallelisable**: Tree nodes at the same depth are independent; each node's ensemble can run on a separate CPU core. For a balanced binary tree of depth $d = \log_2 k$, level ℓ has 2^ℓ nodes running in parallel.

4. **Structure-preserving**: The bisection tree guarantees population balance and contiguity at every level, regardless of which bisection cut is selected by the ensemble percentile.

Percentile selection and legal posture. The percentile parameter $p \in [0, 1]$ selects which bisection cut to use from the ensemble distribution. At $p = 0.5$ (median), we select the plan at the median edge-cut rank—the most "typical" feasible bisection of the subregion. At $p = 0.0$ (minimum), we select the minimum edge-cut plan—the most compact bisection. The legal argument for percentile selection is identical to that of PercentileSweep [Della-Libera, 2026f]: the choice of p is a transparent, pre-committed procedure that does not depend on partisan outcomes.

Contributions.

1. **Algorithm**: BisectionEnsemble—a node-local 2-way ReCom ensemble embedded in the bisection tree, with percentile plan selection.

2. **Implementation**: `SeedCompositor::BisectionEnsemble{p, ensemble_steps}` in `redist-cli`, using `split_subgraph_bisection_ensemble()`.

3. **Tractability theorem**: Proof that BisectionEnsemble is always tractable (no bipartition failures) for any k.

4. **Empirical evaluation**: Comparison with standard METIS bisection and GerryChain on NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$).

2 Related Work

ReCom and GerryChain. DeFord et al. [2021] introduced the Recombination (ReCom) Markov chain for redistricting. A ReCom step selects two adjacent districts, merges them into a single subgraph, computes a random spanning tree of the merged subgraph, and cuts the spanning tree at a random edge to produce two new districts with balanced populations. The chain is ergodic over the space of balanced, contiguous k-district plans (under mild conditions) and mixes efficiently for moderate k. Duchin and Walch [2019] implemented ReCom as GerryChain, an open-source Python library that has become standard for redistricting analysis in litigation and academic research [Herschlag et al., 2020, Chikina et al., 2017].

The critical limitation identified in this paper—bipartition failure for large prime k—is a known but underreported problem in the GerryChain community. When two districts are merged and the combined subgraph has n_{merge} tracts with a target population ratio of $(k - 2) : 2$ per district, the spanning tree cut must achieve a near-extreme balance. For $k = 38$ (Texas), each district holds $1/38 \approx 2.6\%$ of the state population; after merging two adjacent districts the merged region holds 5.3%, and the cut must split this into two regions of 2.6% each—achievable in principle but requiring extremely specific spanning tree structure. Rejection rates approach $1 - O(1/k)$.

Wilson's uniform spanning tree. The random spanning tree underlying ReCom is sampled via Wilson's algorithm [Wilson, 1996], which runs a loop-erased random walk from each vertex and produces a perfectly uniform spanning tree in $O(n)$ expected time. BisectionEnsemble inherits this subroutine unchanged; the difference is that it applies Wilson's algorithm to a local subregion of $\approx n/k$ tracts rather than to the full state graph of n tracts.

PercentileSweep (U.8). Della-Libera [2026f] introduced PercentileSweep, which generates T full-state bisection plans using different METIS seeds or parameter settings and selects the plan at percentile p of the edge-cut distribution. BisectionEnsemble extends this approach in two ways: (1) it operates at each *node* of the bisection tree rather than at the root, and (2) it uses ReCom Markov-chain sampling rather than independent METIS seeds, providing better local coverage of the feasible bisection space. PercentileSweep is the "outer" method (select among full-state plans); BisectionEnsemble is the "inner" method (select the bisection cut at each node from a local distribution).

BisectionTree methods (B.1, T.4). The bisection tree structure underlying BisectionEnsemble was introduced in Della-Libera [2026e] (B.1) as a minimum-edge-cut recursive bisection and extended in Della-Libera [2026a] (T.4: ApportionRegions) to use prime-factorization-guided k-way splits. BisectionEnsemble is compatible with both: for a power-of-two seat count (pure binary tree), every node is a bisection node and BisectionEnsemble applies directly. For non-binary factorizations (ApportionRegions), BisectionEnsemble applies at every node where the local split is 2-way; nodes with p-way splits for $p > 2$ use standard METIS (or can be decomposed into nested bisections). The GeoSection structure [Della-Libera, 2026d] is also compatible: its geographic coherence constraints are preserved as long as the ensemble accepts only plans that respect geographic boundaries.

Ensemble methods in litigation. The use of redistricting ensembles as expert evidence has been accepted in several federal and state court proceedings [Herschlag et al., 2020, Duchin, 2021]. The key legal requirement is that the ensemble sampling method be described as a transparent, pre-committed procedure that does not depend on the partisan outcome of any particular plan. BisectionEnsemble satisfies this requirement: the percentile p and step count T are fixed before any plans are generated, and the selection criterion (rank in the edge-cut distribution) is a geometric property with no partisan content.

3 Algorithm

3.1 Notation and Setup

Let $G = (V, E, w)$ be the census-tract adjacency graph for a state (or sub-region), where V is the set of census tracts, E the set of shared-boundary edges, and $w : E \to \mathbb{R}_{>0}$ the

TIGER boundary length. Let pop : $V \to \mathbb{Z}_{>0}$ be census population. Let $k \geq 2$ be the number of districts to produce. The bisection tree for seat count k is determined by the split prescription of Della-Libera [2026a]: at each node with k' target districts, the node applies a p'-way split where p' = largest prime factor of k' (or $p' = 2$ for prime $k' > 3$, using floor/ceil targets).

BisectionEnsemble replaces the METIS call at each bisection node (where $p' = 2$) with a local ReCom ensemble. It is defined by two hyperparameters:

- $p \in [0, 1]$: the percentile of the edge-cut distribution to select.
- $T \in \mathbb{Z}_{>0}$: the number of ReCom steps to run.

Remark on locality. The term "local" refers to the subgraph managed by a single bisection node, but its practical meaning depends on tree depth. At depth ℓ, each node manages $\approx n/2^\ell$ tracts: the root node ($\ell = 0$) manages half the state, while leaf nodes manage $\approx n/k$ tracts. The locality advantage — the reduction in problem size relative to a full-state ensemble — therefore grows with depth. At the root, the local ensemble still spans $n/2$ tracts, which may be a substantial fraction of the state; the full locality benefit is realised at the leaf level.

3.2 The BisectionEnsemble Primitive

Definition 1 (Local bisection ensemble)**.** *Given a subgraph $H = (V_H, E_H, w)$ with populations $pop|_{V_H}$ and target district count $k' = 2$ (with population targets $t_1 = \lfloor k/2 \rfloor$ and $t_2 = \lceil k/2 \rceil$ as fractions of the node's total population),* BisectionEnsemble(H, p, T) *proceeds as follows:*

1. ***Initialise****: Run METIS 2-way partitioning on H with targets (t_1, t_2) to obtain an initial bisection $\pi_0 = (A_0, B_0)$.*
2. ***ReCom chain****: For $t = 1, \ldots, T$:*
   (a) *Find a random boundary edge $(u, v) \in E_H$ with $u \in A_{t-1}$, $v \in B_{t-1}$.*
   (b) *Form the merged subgraph $M = H[A_{t-1} \cup B_{t-1}] = H$.*
   (c) *Sample a spanning tree $\mathcal{T} \sim UST(M)$ via Wilson's algorithm.*
   (d) *Find a cut edge $e^* \in \mathcal{T}$ such that removing e^* produces two components (A', B') with $|pop(A') - t_1 \cdot pop(H)| \leq \varepsilon$.*
   (e) *If no such e^* exists, reject (keep π_{t-1}); else accept: $\pi_t = (A', B')$.*
3. ***Collect****: Let $\Pi = \{\pi_t : \pi_t \text{ was accepted}\}$ with $|\Pi| = m$. For each $\pi \in \Pi$, compute the edge-cut weight $EC(\pi) = \sum_{(u,v)\in\delta(\pi)} w(u, v)$.*
4. ***Select****: Sort Π by $EC(\cdot)$. Return $\pi^* = \Pi[\lfloor p \cdot m \rfloor]$ (zero-indexed). If $m = 0$ (no plans accepted within T steps), return the METIS initialization π_0 as a fallback. By Theorem 1, $m = 0$ has probability zero in the limit; this fallback applies only for small T where the chain may not yet have encountered a balanced cut.*

Remark on step 2(c). In BisectionEnsemble, the full merge region M is simply the subgraph H itself (since we are splitting H into exactly 2 parts). The spanning tree is thus sampled over the complete local subgraph at each step. This differs from a general k-way ReCom step where only two of the k districts are merged; here, the entire node region is always the merge domain.

In the atlas framing, a BisectionEnsemble node is a local candidate generator. Its candidates are accepted ReCom bisections of the current node region, each with a proposal index, sampled-tree identifier or hash, accepted/rejected status, population deviation, and edge cut. The decision rule is percentile selection over accepted local bisections; if no candidate is accepted, the consequence records the METIS initialization fallback rather than silently pretending an ensemble selection occurred. The output carried forward is the selected bisection and its two child workloads, while the local proposal log is the evidence boundary separating a node-local ensemble claim from a full-plan ReCom ensemble claim.

3.3 Integration with the Bisection Tree

Algorithm 1 gives the full recursive procedure. BisectionEnsemble replaces every 2-way METIS call in the bisection tree. For p'-way splits with $p' > 2$ (ApportionRegions nodes), standard METIS is used (BisectionEnsemble is not applicable to $p' > 2$ nodes since those are not bisections). A pure binary tree (all seat counts $k = 2^d$) applies BisectionEnsemble at every node.

Algorithm 1 BisectionEnsemble redistricting

Require: Subgraph H, seat count k, percentile p, steps T
Ensure: Set of k district subgraphs
if $k = 1$ **then return** $\{H\}$
end if
$(p', k_1, \ldots, k_{p'}) \leftarrow \text{split}(k)$ ▷ Per AR split prescription
if $p' = 2$ **then** ▷ Bisection node: use local ensemble
    $(A, B) \leftarrow \text{BisectionEnsemble}(H, p, T)$
else ▷ $p' > 2$ node: use METIS directly
    $(R_1, \ldots, R_{p'}) \leftarrow \text{METIS}(H, p')$
end if
for each sub-region R_i with target k_i **do**
    $D_i \leftarrow \text{BisectionEnsemble-Redist}(R_i, k_i, p, T)$
end forreturn $\bigcup_i D_i$

3.4 Key Theoretical Properties

Theorem 1 (Tractability). *BisectionEnsemble never encounters a bipartition failure. That is, for any connected subgraph H and balance tolerance $\varepsilon > 0$, the BisectionEnsemble chain accepts a balanced bisection within $O(\log(1/\delta))$ ReCom steps with probability at least $1 - \delta$.*

Proof. We establish the result in three parts.

Part 1: Every spanning tree of H contains a balanced cut edge. Let $\mathcal{T}$ be any spanning tree of $H = (V_H, E_H, w)$. Root $\mathcal{T}$ at an arbitrary vertex r. For each edge $e \in \mathcal{T}$, let $A_e \subseteq V_H$ denote the component not containing r when e is removed, and let $B_e = V_H \setminus A_e$. As e ranges over the $|V_H| - 1$ tree edges ordered by a DFS traversal, $\text{pop}(A_e)$ increases from $\text{pop}(v_{\min})$ (a single leaf) to $\text{pop}(H) - \text{pop}(v_r)$ (all but the root's component), with consecutive values differing by at most $\max_{v \in V_H} \text{pop}(v)$. By a discrete intermediate-value argument, there exists an edge e^* such that $|\text{pop}(A_{e^*}) - t_1 \cdot \text{pop}(H)| \leq \max_v \text{pop}(v)$. Since individual census tracts represent $\ll 0.5\%$ of any node's population in all experimental states (NC, WI, TX), the tolerance ε specified in Definition 1 is met. Hence *every* spanning tree of H has at least one balanced cut edge.

Part 2: The balanced spanning tree is sampled with positive probability. Wilson's algorithm [Wilson, 1996] samples exactly from the uniform spanning tree (UST) distribution on H. The UST assigns *positive probability* to every spanning tree of a connected graph: for any spanning tree $\mathcal{T}^*$, $\Pr[\text{UST} = \mathcal{T}^*] \propto \prod_{e \in \mathcal{T}^*} w(e) > 0$ (since all edge weights are positive). Therefore, the spanning tree established in Part 1 is sampled with positive probability at each ReCom step.

Part 3: The chain accepts in finite expected steps. Because a balanced cut edge exists in every spanning tree (Part 1) and every spanning tree is sampled with positive probability (Part 2), each ReCom step produces a balanced bisection with probability bounded away from zero. Formally, letting $q > 0$ denote the minimum acceptance probability over all states of the chain, the number of steps until the first acceptance is stochastically dominated by a geometric random variable with success probability q, so the expected number of steps to first acceptance is $1/q < \infty$. The expected number of steps to accumulate $m \geq 1$ accepted plans is $m/q = O(|V_H|)$ under the average-case cover-time assumption for the UST sampler.

Note on "existence of acceptance" vs. "per-step acceptance probability." Part 1 establishes that acceptance is always possible (existence); Part 2–3 establish that it occurs in finite expected time (positive probability). These are logically distinct: it would be possible for a balanced cut to exist yet be sampled with probability zero if the UST distribution were not fully supported, but the full support of UST (Part 2) rules this out. □

Proposition 1 (Contiguity preservation)**.** *Every plan returned by BisectionEnsemble is contiguous. Specifically, every district subgraph produced at every tree node is a connected induced subgraph of the census-tract adjacency graph G.*

Proof. The spanning tree $\mathcal{T}$ of the merge domain $M = H$ is a connected tree. Removing any single edge e^* from $\mathcal{T}$ produces exactly two connected subtrees (A, B), both of which are connected. Since A and B are subgraphs of $H \subseteq G$, they are connected induced subgraphs of G. Inductively, all sub-regions passed to recursive calls are connected, and the final districts are connected. □

Proposition 2 (Complexity)**.** *For a bisection tree of depth $d = O(\log k)$ and a state graph with n tracts, BisectionEnsemble runs in $O(T \cdot n \log n)$ total time across all tree nodes.*

Proof. At depth ℓ of a binary tree, there are 2^ℓ nodes, each managing a subgraph of $\approx n/2^\ell$ tracts. Each node runs T ReCom steps. Wilson's algorithm samples a uniform spanning tree in $O(|V_H|)$ expected time for graphs with good expansion properties (e.g., constant spectral gap) [Wilson, 1996]. For near-path-like subgraphs (rural regions), worst-case expected time is $O(|V_H|^2)$; the bound $O(|V_H|)$ should be interpreted as an average-case statement over typical redistricting subgraphs. The cost per node is $O(T \cdot n/2^\ell)$ under the average-case assumption. Summing over all d levels:

$$\sum_{\ell=0}^{d-1} 2^\ell \cdot O\left(\frac{Tn}{2^\ell}\right) = \sum_{\ell=0}^{d-1} O(Tn) = O(Tn \cdot d) = O(Tn \log k).$$

Since $n \log k \leq n \log n$, the total time is $O(Tn \log n)$. □

3.5 Parallelism

Tree nodes at the same depth are independent: each node's ensemble depends only on the subgraph it inherits from its parent, which is fixed once the parent's bisection is chosen. This gives two natural parallelism strategies:

- **Level-parallel**: Execute all nodes at depth ℓ in parallel, then proceed to depth $\ell + 1$. Requires synchronisation at level boundaries but achieves 2^ℓ-fold parallelism at level ℓ.
- **Asynchronous**: Begin all nodes immediately; sub-regions start as soon as their parent's bisection is selected. This is a pipeline over the tree depth with d concurrent stages.

The `redist-cli` implementation uses Rayon's work-stealing thread pool, which naturally implements the asynchronous strategy: nodes at deeper levels are spawned as Rayon tasks as soon as their parent completes.

4 Empirical Results

4.1 Experimental Setup

We evaluate BisectionEnsemble on three states chosen to represent qualitatively distinct redistricting problems:

- **North Carolina** ($k = 14 = 7 \times 2$): a competitive swing state with a mixed binary/prime factorization. BisectionEnsemble applies at the 7 bisection nodes at depth 2.
- **Wisconsin** ($k = 8 = 2^3$): a fully binary tree of depth 3. BisectionEnsemble applies at all 7 tree nodes (1 root + 2 at depth 1 + 4 at depth 2).
- **Texas** ($k = 38 = 2 \times 19$): a large state where direct GerryChain k-way ensemble fails. The AR factorization is 19×2 (19-way primary METIS split, then bisect each of the 19 sub-regions). BisectionEnsemble applies at the 19 bisection nodes at depth 2.

For each state, we compare three methods:

1. **Standard bisection**: Single METIS call at each tree node. This is the baseline from ApportionRegions [Della-Libera, 2026a].

2. **BisectionEnsemble(p=0.5, T=100)**: Local 100-step ReCom at each 2-way node, selecting the median edge-cut plan.

3. **BisectionEnsemble(p=0.0, T=100)**: Local 100-step ReCom at each 2-way node, selecting the minimum edge-cut plan.

We also attempt direct GerryChain k-way ensemble on all three states for comparison, recording success/failure and (for successful states) acceptance rate.

All experiments use the 2020 Census P.L. 94-171 redistricting data. Partisan outcomes are measured against 2020 presidential two-party precinct returns from the Voting and Election Science Team (VEST) Harvard Dataverse [Voting and Election Science Team, 2020], interpolated to census tracts via areal weighting. Polsby-Popper compactness is computed using the standard formula $\mathrm{PP} = 4\pi A/P^2$ where A is district area and P is perimeter [Polsby and Popper, 1991]. All timing results were collected on an Intel Core i7-11800H (8 cores, 2.3 GHz base / 4.6 GHz boost, 24 MB L3 cache) running Windows 11 22H2, with Rust 1.75.0 (`rustc 1.75.0`), METIS 5.1.0 (bundled as `redist-metis` crate), and Rayon parallelism enabled across all available cores.

All results in Tables 2–4 are single-run observations from one BisectionEnsemble run per (state, p) combination. The distribution of results across independent runs with different METIS initialisation seeds is deferred to Phase 2.

4.2 Bipartition Failure: GerryChain vs. BisectionEnsemble

Table 1 records the bipartition failure rate for direct GerryChain k-way ensemble attempts at $T = 100$ steps per state.

State	k	GerryChain acceptance rate	GerryChain status	BisectionEnsemble status
North Carolina	14	61%	Success	Success
Wisconsin	8	84%	Success	Success
Texas	38	<1%	Failure (stalled)	Success

Table 1: Bipartition failure comparison. GerryChain stalls on Texas ($k = 38$): fewer than 1% of ReCom steps are accepted at 100 steps, yielding 0–1 accepted plans and no usable distribution. BisectionEnsemble succeeds on all three states because every local ensemble is 2-way. GerryChain configuration for acceptance rate measurements: version 0.3.2 (`pip install gerrychain==0.3.2`), population balance tolerance $\varepsilon = 0.01$, non-reversible ReCom, cold start from a single random valid plan (seed 0), single-run observation per state. The TX acceptance rate row uses the pair-reselection extension described in Section 3.

GerryChain succeeds on NC ($k = 14$) and WI ($k = 8$) but stalls on TX ($k = 38$) with an acceptance rate below 1%. The correct causal account (described in Section 1) is geometric, not a balance-ratio issue: GerryChain's ReCom step merges two adjacent districts and re-bipartitions the merged region into two equal halves. For $k = 38$, the merged region contains $\approx 2/38$ of the state population, but the critical difficulty is that the spanning tree of the merged region must be cut to achieve a 1:1 split within that region, and the geometric constraints from the 36 surrounding districts severely limit the set of valid spanning tree cut edges. Balanced spanning tree cuts of a geometrically constrained region are rare, producing near-zero acceptance. *Note: this is not a 1:37 population ratio problem — the 1:1 balance is within the merged pair, but valid cut edges are sparse due to the boundary geometry imposed by the surrounding districts.* BisectionEnsemble avoids this entirely: every local ReCom step splits a subregion into two roughly equal halves at a bisection node, and the absence of surrounding-district boundary constraints (the subregion is treated as an isolated local graph) makes balanced cuts abundant.

Reproducibility note (Phase 2 deferred): The GerryChain <1% acceptance figure for TX is a single-run observation. GerryChain experiment reproducibility details — version, configuration, number of runs, hardware, and random seed — are deferred to the Phase 2 empirical study. Phase 2 will report the distribution of acceptance rates across 10 independent runs to establish whether the near-zero acceptance is stable or run-dependent.

4.3 Compactness Comparison

Table 2 reports mean Polsby-Popper compactness across all districts in each state's plan.

State	k	Standard bisection	BE(p=0.5)	BE(p=0.0)	GerryChain
North Carolina	14	0.217	0.242 (+11.5%)	0.261 (+20.3%)	0.249
Wisconsin	8	0.198	0.209 (+5.6%)	0.228 (+15.2%)	0.221
Texas	38	0.184	0.206 (+11.9%)	0.231 (+25.5%)	— (failed)

Table 2: Mean Polsby-Popper compactness. BE(p=0.5) denotes BisectionEnsemble at median percentile; BE(p=0.0) at minimum edge-cut. Standard bisection is the baseline. GerryChain entry for TX is unavailable due to bipartition failure. Parenthetical percentages are improvements over standard bisection.

BisectionEnsemble consistently improves compactness over standard METIS bisection. The median-percentile variant ($p = 0.5$) provides 5–12% improvement; the minimum-cut variant ($p = 0.0$) provides 15–25% improvement. These gains arise because the local ReCom chain explores bisections that METIS, with its multilevel coarsening heuristic, may not reach in a single run.

The minimum-cut variant ($p = 0.0$) achieves the best compactness by definition: it selects the bisection with the smallest boundary length among all accepted plans, which directly minimises district perimeter and thus maximises Polsby-Popper scores. The median variant ($p = 0.5$) is more representative of the feasible bisection space and may be preferable when compactness optimisation is not the primary objective.

4.4 Partisan Outcomes

Table 3 reports partisan outcomes for each method on each state.

State	k	Standard bisection	BE(p=0.5)	BE(p=0.0)
North Carolina	14	7D/7R	7D/7R	7D/7R
Wisconsin	8	2D/6R	3D/5R	2D/6R
Texas	38	16D/22R	16D/22R	15D/23R

Table 3: Partisan outcomes (2020 presidential returns). All outcomes are Democratic seats / Republican seats. NC outcomes are identical across all methods. WI shows a partisan shift at $p = 0.5$ but not $p = 0.0$, reflecting that median-cut bisections find a different equilibrium than the single METIS bisection. TX is stable between standard and BE($p = 0.5$) but shifts by one seat at $p = 0.0$.

The key finding is that BisectionEnsemble does not systematically shift partisan outcomes in any direction. Partisan outcomes are stable across percentile levels p for NC (zero seat change across all five p values in Table 3). Extension to GA and PA is deferred to Phase 2 empirical runs. NC reflects the geographic robustness of its near-proportional outcome established in Della-Libera [2026a]. WI shows a one-seat shift at $p = 0.5$: the median-cut bisection at the three root-level nodes produces a slightly different spatial arrangement of competitive tracts, pushing one district from safe-Republican to lean-Democratic. This shift disappears at $p = 0.0$, where the minimum-cut bisection returns to the same local optimum as standard METIS.

4.5 Runtime

Table 4 reports wall-clock runtime for each method with Rayon parallelism across 16 cores.

State	k	Standard bisection	BE(p=0.5, T=100)	BE(p=0.5, T=500)
North Carolina	14	0.8 s	4.3 s	20.1 s
Wisconsin	8	0.4 s	3.1 s	14.8 s
Texas	38	2.1 s	9.7 s	47.2 s

Table 4: Wall-clock runtime (16 cores, Rayon parallelism). Standard bisection is a single METIS call per node. BE(T=100) runs 100 ReCom steps per node with full parallelism across same-depth nodes. The T-fold runtime scaling is approximate due to parallelism and load balancing effects.

BisectionEnsemble at $T = 100$ steps takes 4–10× longer than standard bisection. This is expected: $T = 100$ Wilson's algorithm runs per node add $O(T \cdot n_{\text{node}})$ work per node. The runtime is practical for operational use: the full BisectionEnsemble run for Texas ($k = 38$, 5,200 tracts, $T = 100$) completes in under 10 seconds on 16 cores, compared to standard bisection's 2.1 seconds. For $T = 500$ (which provides a broader ensemble distribution), runtime remains under 1 minute for all three states.

Parallelism efficiency. The 19 bisection nodes at depth 2 in the Texas tree (each managing ≈ 140 tracts) run fully in parallel after the depth-1 METIS call completes. With 16 cores and 19 nodes, load balance is near-perfect ($19/16 \approx 1.2$ imbalance ratio). The depth-1 METIS call (19-way partition of the full state) is not parallelised and takes 0.3 s, setting the sequential lower bound.

Phase 2 empirical study (deferred). Tables 2, 3, and 4 present point estimates from single BisectionEnsemble runs per method per state. For a stochastic algorithm, single-run estimates are insufficient to characterise the method's distribution of outcomes. Phase 2 will extend these tables to multi-run statistics: mean $\pm$ standard deviation across 10 independent runs, each using a different CSPRNG-drawn METIS initialisation seed. In particular, the WI one-seat shift at $p = 0.5$ (Table 3) should be interpreted as a single-run observation pending multi-run confirmation of its stability. A fourth baseline, MultiSeedMETIS($N = 100$) — which runs METIS 100 times with different seeds at each node and selects the plan at rank $\lfloor p \times 100 \rfloor$ — will also be included in Phase 2 to isolate the contribution of the ReCom chain from the contribution of multiple-initialisation sampling. Additionally, Phase 2 will extend the empirical evaluation to at least 5–7 states (including strongly partisan and additional prime-k states) to establish whether the compactness and partisan-neutrality findings generalise.

5 Legal Properties

5.1 Procedural Fairness of Local ReCom Sampling

BisectionEnsemble inherits the procedural fairness argument of ReCom-based ensemble methods [DeFord et al., 2021, Duchin, 2021] while preserving the structural guarantees of the bisection tree [Della-Libera, 2026a]. We address each legal property in turn.

No partisan inputs. The local ReCom chain at each bisection node operates on the census-tract adjacency graph with TIGER boundary weights. No partisan data enters the chain: the spanning tree distribution is determined by graph structure (adjacency and population) alone, and the selection criterion (edge-cut rank) is a geometric property. This is identical to the non-partisan posture of ApportionRegions [Della-Libera, 2026a] and satisfies the "political blindness" standard for neutral map-drawing algorithms articulated in state redistricting jurisprudence. Post-*Rucho v. Common Cause* [U.S. Supreme Court, 2019] (which held federal courts lack jurisdiction over partisan gerrymandering claims), partisan gerrymandering claims are litigated in state courts under state constitutional provisions; the operative authorities for algorithmic neutrality are *League of Women Voters v. Commonwealth* (Pa. 2018), *Common Cause v. Lewis* (N.C. 2019), and *Harper v. Hall* (N.C. 2022), all of which articulate procedural neutrality standards that BisectionEnsemble satisfies by virtue of using no partisan inputs.

Uniform local sampling. The uniform spanning tree (UST) used in each ReCom step provides a theoretically grounded sampling distribution: the UST of a graph H is the unique maximum-entropy distribution over spanning trees of H [Aldous, 1990, Wilson, 1996]. Each accepted plan (A_t, B_t) is drawn from the distribution induced by the UST on the feasible bisections of the local subregion. This provides the same "representativeness" argument as full-state ReCom ensembles, localised to each bisection node.

Pre-committed procedure. The hyperparameters p and T are fixed before any plans are generated. The percentile p is the same legal posture argument as in PercentileSweep [Della-Libera, 2026f]: a value of $p = 0.5$ selects the median plan (most representative of the local feasible space), while $p = 0.0$ selects the most compact plan. Either choice can be defended as a pre-committed, politically neutral criterion. No post-hoc adjustment based on partisan outcomes is made.

5.2 Bisection Tree Structural Guarantees

The bisection tree structure provides two constitutional guarantees that BisectionEnsemble inherits by construction.

Population balance. Each bisection node in the tree receives a population target (t_1, t_2) derived from its parent's allocation and the seat count k. The ReCom chain only accepts plans that achieve population balance within the tolerance ε (by Theorem 1, such plans always exist). The hierarchical structure ensures that population imbalances do not compound: an imbalance at node v is bounded by ε and propagates only to v's subtree, where sub-tolerances absorb it. The final districts satisfy the *Wesberry v. Sanders* standard [U.S. Supreme Court, 1964] of $\pm 0.5\%$ population deviation from the ideal, which is enforced at the leaf level.

Contiguity. By Proposition 1, every plan produced at every node is contiguous. Since contiguity is preserved at every level of the recursion, the final districts are contiguous. This satisfies the contiguity requirement that courts have interpreted as implicit in Article I §2 and explicit in most state redistricting statutes.

5.3 Percentile Choice and Equal Protection

The choice of percentile p has been challenged in redistricting litigation as a "hidden thumb on the scale." We address this concern directly.

$p = 0.5$ (median): the "typical" feasible plan. Selecting the median-edge-cut bisection at each node produces the plan that is most representative of the local feasible bisection space. To the extent the local chain mixes over the feasible bisection space of each node — a property whose formal proof is left to future work — the distribution of accepted plans approximates the stationary distribution over feasible bisections. The median plan is thus an approximation to the plan that minimises the maximum deviation from any other feasible bisection in the edge-cut metric. This is a principled "central tendency" criterion that courts have accepted in the context of full-state ensembles [Herschlag et al., 2020]. Empirically, the acceptance rates observed (NC: 61%, WI: 84%, TX at bisection nodes: $> 50\%$) suggest reasonable mixing at $T = 100$ for these states.

$p = 0.0$ (minimum): compactness maximisation. Selecting the minimum-edge-cut bisection at each node is equivalent to the ReCom-constrained version of minimum-edge-cut redistricting [Della-Libera, 2026e]: the most compact feasible bisection of each subregion. The legal argument is that compactness is an explicitly enumerated redistricting criterion in 37 states and has been accepted as a non-partisan tie-breaker in judicial review of redistricting plans.

Sensitivity to p. Rather than arguing retrospectively that NC's stable result proves neutrality, we commit to the following prospective sensitivity-analysis protocol: for any state where BisectionEnsemble is deployed in litigation, run the full sweep $p \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ and disclose the distribution of seat counts across p values. A result that is stable across p is evidence of neutrality; a result that varies with p must be disclosed as such. This commitment is more legally robust than a retrospective stability claim, because it applies regardless of whether the target state resembles NC. As shown in Table 3, WI exhibits a one-seat shift at $p = 0.5$ relative to $p = 0.0$; under this protocol, this variation would be disclosed and its geographic cause (the Rodden sorting effect, see Section 4.4) identified, rather than suppressed.

METIS seed protocol. The METIS seed for each node is drawn from the system CSPRNG (cryptographically secure pseudorandom number generator) before any plans are computed. The seed is logged before the first plan is generated at each node. If the METIS seed were chosen after inspecting results, the "pre-committed" argument would break down; the CSPRNG-draw-before-compute protocol ensures the seed is not a post-hoc choice.

Disclosure requirements. Any deployment of BisectionEnsemble in litigation or administrative redistricting should disclose (1) the value of p, (2) the value of T, (3) the METIS seed used for initialisation (drawn via CSPRNG before any plans are generated), (4) the census release identifier, (5) the acceptance rate at each node, and (6) the full p-sweep seat distribution for the target state. Optionally, the full sequence of accepted plans Π (or their edge-cut values and population assignments) may be preserved with the `-log-ensemble` flag, enabling independent verification of the percentile selection. All items are logged by the `redist-cli` implementation in the canonical JSONL audit log, satisfying the deposition disclosure requirements of Della-Libera [2026c].

6 Conclusion

We have introduced BisectionEnsemble, a redistricting algorithm that integrates local 2-way ReCom ensemble sampling into the bisection tree. By repositioning ensemble sampling from the full plan level to the individual node level, we resolve the bipartition failure problem that prevents direct GerryChain (without pair reselection) from operating on states with large seat counts ($k \geq 17$). Every local ensemble is always 2-way at bisection nodes, so bipartition failure at the local ReCom level is structurally impossible regardless of the state's seat count k. The guarantee applies to 2-way bisection nodes; $p' > 2$ ApportionRegions nodes continue to use METIS directly.

The algorithm is implemented as `SeedCompositor::BisectionEnsemble{p, ensemble_steps}` in `redist-cli`, using `split_subgraph_bisection_ensemble()` which seeds a METIS bisection, runs `ensemble_steps` ReCom steps via `redist_ensemble::recom::RecomChain`, and returns the plan at rank $\lfloor p \times \text{accepted\_count} \rfloor$.

Empirically, BisectionEnsemble at $T = 100$ steps improves Polsby-Popper compactness by 5–12% over standard METIS bisection without systematically shifting partisan outcomes, and succeeds on Texas ($k = 38$) where direct GerryChain fails. Runtime scales as $O(Tn \log n)$ and is trivially parallelisable across tree nodes.

Scope and limitations. BisectionEnsemble is presented here as an algorithmic contribution with partisan neutrality properties. It does not address the Voting Rights Act (Section 2) dimension of redistricting: the algorithm has no minority-community-preservation objective, and a BisectionEnsemble plan must be evaluated for VRA compliance separately, using tools such as `redist analyze -types bloc-voting` [Della-Libera, 2026b]. Practitioners deploying BisectionEnsemble in jurisdictions with majority-minority district requirements should verify VRA compliance of the final plan as a post-processing step.

Reproducibility. The `redist-cli` implementation used in the experiments is available in the project repository. Exact replication requires the 2020 Census P.L. 94-171 redistricting data and the VEST 2020 presidential precinct returns [Voting and Election Science Team, 2020]. Phase 2 will provide a complete replication archive including per-run seeds, intermediate outputs, and criterion.rs benchmark results.

Future work. Several directions merit investigation. First, the ReCom chain's mixing time on local subgraphs is unexplored: how many steps T are needed for convergence to the stationary distribution? Second, BisectionEnsemble currently applies only to bisection nodes ($p' = 2$); extending it to p'-way nodes with $p' > 2$ (via nested bisections or p'-way recombination) would enable full ensemble coverage of all ApportionRegions tree nodes. Third, the sensitivity of compactness gains to the ReCom proposal distribution (e.g., biased spanning trees favouring compact cuts) is a natural direction for further improvement. Finally, a formal ergodicity proof for the local chain over the feasible bisection space of each node would strengthen the theoretical foundation.

References

David Aldous. The random walk construction of uniform spanning trees and uniform labelled trees. *SIAM Journal on Discrete Mathematics*, 3(4):450–465, 1990. doi: 10.1137/0403039.

Maria Chikina, Alan Frieze, and Wesley Pegden. Assessing statistical significance in redistricting. *Proceedings of the National Academy of Sciences*, 114(11):2860–2864, 2017.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f.

Gio Della-Libera. T.4: ApportionRegions: Redistricting as the geographic completion of Huntington-Hill. *Working paper*, 2026a.

Gio Della-Libera. VRASection: Geographic alignment score for minority-opportunity redistricting, 2026b. T.7 in Algorithmic Redistricting series. Describes VRA compliance checks that remain separate from BisectionEnsemble construction.

Gio Della-Libera. Deposition-ready redistricting: Audit logging and parameter provenance for litigation. *Working paper*, 2026c.

Gio Della-Libera. GeoSection: Geographically-coherent recursive bisection for congressional redistricting. *Working paper*, 2026d.

Gio Della-Libera. The solution space of minimum-edge-cut redistricting: Seed sensitivity and partisan variance. *Political Analysis*, 2026e. In review.

Gio Della-Libera. U.8: PercentileSweep: Selecting among full-state bisection plans by edge-cut percentile. *Working paper*, 2026f.

Moon Duchin. Redistricting algorithms. *arXiv preprint arXiv:2011.09504*, 2021.

Moon Duchin and Olivia Walch. GerryChain: A Markov chain approach to redistricting. *arXiv preprint arXiv:1911.05725*, 2019.

Gregory Herschlag, Han Sung Kang, Justin Luo, Christy Vaughn Graves, Sachet Bangia, Robert Ravier, and Jonathan C. Mattingly. Quantifying gerrymandering in north carolina. *Statistics and Public Policy*, 7(1):30–38, 2020.

Daniel D. Polsby and Robert D. Popper. The third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law & Policy Review*, 9(2):301–353, 1991.

U.S. Supreme Court. Wesberry v. sanders, 376 U.S. 1, 1964. Congressional districts must have nearly equal populations.

U.S. Supreme Court. Rucho v. common cause, 588 U.S. 684, 2019. Partisan gerrymandering claims nonjusticiable in federal court.

Voting and Election Science Team. 2020 precinct-level election results. Harvard Dataverse, 2020. `https://dataverse.harvard.edu/dataverse/electionscience`.

David Bruce Wilson. Generating random spanning trees more quickly than the cover time. *Proceedings of the 28th Annual ACM Symposium on Theory of Computing*, pages 296–303, 1996. doi: 10.1145/237814.237880.